Table 1.1. Many of these discoveries and their practical consequences are discussed in the subsequent text.

1.5. NUCLEAR POWER

The peaceful use of the nuclear fission energy to produce electricity was first successfully employed in 1951 by the EBR-1 (Experimental Breeder Reactor) in the US. This was a fast breeder reactor using enriched uranium as fuel and cooled by a NaK eutectic melt. Originally it produced enough electricity to power four 200 W light bulbs and subsequently enough to power its own building. In 1955 the EBR-1 suffered a partial melt down and was finally deactivated in 1964. Admittedly the EBR-I produced the first electricity it was not really connected to the grid. This was achieved by the nearby plant, BORAX-III, which in 1955 made the nearby city of Arco to be the first city solely powered by nuclear energy.

Similar activities were performed in the Soviet Union where an existing graphite moderated channel-type plutonium production reactor was modified for heat and electricity generation by introducing a heat exchange system. It became the world's first thermal neutron nuclear powered electricity generator in 1954. The reactor was called AM-1 (Atom Mirny — peaceful atom) with a capacity of 30 MW_{th} or 5 MW_e. AM-1 produced electricity until 1959 and was used as a research facility and for production of radioisotopes until 2002. As seen above the general construction was close to the later infamous Chernobyl reactor RBMK (reaktor bolshoi moshchnosty kanalny — high power channel reactor).

The pressurised water reactor (PWR) history start with the use in naval propulsion. The Mark 1 prototype naval reactor became critical in 1953 which led to the first nuclear-powered submarine, *USS Nautilus*, was launched in 1954. This reactor led to the building in 1957 of the 60 MWe Shippingport demonstration PWR reactor in Pennsylvania. This reactor operated until 1982.

A boiling water reactor (BWR) prototype, Vallecitos, was operated from 1957 to 1963 but the first electricity producing one, Dresden-1 of 250 MWe was started in 1960 and was operating until 1978. There reactors also used enriched uranium fuel.

In Canada a different route was selected. Instead of going through the uranium enrichment process they created the CANDU reactor using natural uranium oxide fuel in a heavy water moderated and normal water cooled channel type reactor. The first one was started in 1962.

After these initial steps essentially four reactor systems are in general use: natural water moderated and cooled (PWR/VVER and BWR), heavy water moderated and natural water cooled (CANDU), and graphite moderated and water cooled (RBMK).

Table 1.1 Historical survey of nuclear science

Essential steps in the development of modern science	
~490–430 B.C.	**Empedocles** suggests that everything is made up of four elements: air, earth, water and fire. Every matter can be formed by transmutation between these. (This is principally correct if the four elements are interpreted as being the gaseous, solid and liquid states of matter, and fire interpreted as being energy.)
~460–370 B.C.	**Democritos** proposes that all matter consists of eternal, moving and indestructible atoms, qualitatively alike but differing in size, shape and mass.
1661	**Boyle** writes that the nature is made up of a limited number of substances (elements) which cannot be broken down into simpler ones.
1808	**Dalton**: All chemical compounds (molecules) are combinations of atoms in fixed proportions.
Important steps in the development of nuclear science	
1896	**Becquerel** discovers radiation from uranium (radioactivity). The intensity of the radiation is measured either through its ionization of air or through the scintillations observed when the radiation hits a fluorescent screen.
1896–1905	**Crookes, Becquerel, Rutherford, Soddy, Dorn, Boltwood** *et al.* Radioactive decay is found to be transformation of atoms leading to different radioelements which are genetically connected in radioactive decay series.
1898	**P. and M. Curie** discover polonium and radium; the first radiochemical methods.
1898–1902	**P. Curie, Debierne, Becquerel, Danilos** *et al.* discover that radiation affects chemical substances and causes biological damage.
1900	**Villard** and **Becquerel** propose that -radiation is of electromagnetic nature; finally proven in 1914 by **Rutherford** and **Andrade**.
1900	**Becquerel**: β-rays are identified as electrons.
1902	**M.** and **P. Curie** and **Debierne** isolate first macroscopic amounts of a radioactive element (radium)
1903	**Rutherford** shows α-radiation to be ionized helium atoms.
1905	**Einstein** formulates the law of equivalence between mass and energy.
1907	**Stenbeck** makes the first therapeutic treatment with radium and heals skin cancer.
1911	**Rutherford, Geiger,** and **Marsden** conclude from measurement of the scattering of α-radiation against thin foils that atoms contain a very small positive nucleus.
1912	**Hevesy** and **Paneth**, in the first application of radioactive trace elements, determine the solubility of $PbCrO_4$ using RaD.
1912	**Wilson** develops the cloud chamber, which makes tracks from nuclear particles visible.
1913	**Hess** discovers cosmic radiation.

(*Continued*)

Table 1.1 Historical survey of nuclear science—cont'd

1913	**Fajans** and **Soddy** explain the radioactive decay series by assuming the existence of isotopes. This is proven by **J. J. Thomson** through deflection of neon ions in electromagnetic fields. **Aston** separates the isotopes of neon by gas diffusion.
1913	**N. Bohr** shows that the atomic nucleus is surrounded by electrons in fixed orbitals.
1919	**Rutherford**: first nuclear transformation in the laboratory, $^{4}He + {}^{14}N \rightarrow {}^{17}O + {}^{1}H$.
1919	**Aston** constructs the first practical mass spectrometer and discovers that isotopic weights are not exactly integers.
1921	**Hahn** discovers nuclear isomers: $^{234m}Pa(UX_2) \xrightarrow[1.2\ min]{\gamma} {}^{234}Pa(UZ)$.
1924	**de Broglie** advances the hypothesis that all moving particles have wave properties.
1924	**Lacassagne** and **Lattes** use radioactive trace elements (Po) in biological research.
1925–1927	Important improvements of the Bohr atomic model: Pauli exclusion principle, Schrödinger wave mechanics, Heisenberg uncertainty relationship.
1928	**Geiger** and **Müller** construct the first GM tube for single nuclear particle measurements.
1931	**van de Graaff** develops an electrostatic high voltage generator for accelerating atomic ions to high energies.
1931	**Pauli** postulates a new particle, the neutrino, to be formed in β-decay.
1932	**Cockcroft** and **Walton** develop the high voltage multiplier and use it for the first nuclear transformation in the laboratory with accelerated particles (0.4 MeV $^{1}H + {}^{7}Li \rightarrow 2\ {}^{4}He$).
1932	**Lawrence** and **Livingston** build the first cyclotron.
1932	**Urey** discovers deuterium and obtains isotopic enrichment through evaporation of liquid hydrogen.
1932	**Chadwick** discovers the neutron.
1932	**Andersson** discovers the positron, e^{+} or β^{+}, through investigation of cosmic rays in a cloud chamber.
1933	**Urey** and **Rittenberg** show isotopic effects in chemical reactions.
1934	**Joliot** and **I. Curie** discover artificial radioactivity: $^{4}He + {}^{27}Al \rightarrow {}^{30}P + n$; $^{30}P \xrightarrow[2.5\ min]{\beta^{+}} {}^{30}Si$.
1935	**DeHevesy** develops neutron activation analysis.
1935	**Yukawa** predicts the existence of mesons.
1935	**Weizsäcker** derives the semiempirical mass formulae.
1937	**Neddermeyer** and **Andersson** discover μ-mesons in cosmic radiation using photographic plates.
1938	**Bethe** and **Weizsäcker** propose the first theory for energy production in stars through nuclear fusion: $3\ {}^{4}He \rightarrow {}^{12}C$.
1938	**Hahn** and **Strassman** discover fission products after irradiation of uranium with neutrons.

Table 1.1 Historical survey of nuclear science—cont'd

1938-1939	**Meitner** and **Frisch** interprets the discovery by **Hahn** and **Strassman** as fission of the U-atom by neutrons; this is almost immediately confirmed by several laboratories in Europe and the USA.
1938-1939	**F. Joliot, von Halban, Kowarski** and **F. Perrin** in France apply for patents for nuclear chain reacting energy producing devices and starts building a nuclear reactor; the work is interrupted by the war.
1940	**McMillan, Abelson, Seaborg, Kennedy,** and **Wahl** produce and identify the first transuranium elements, neptunium (Np), and plutonium (Pu), and with **Segré** discover that ^{239}Pu is fissionable.
1940	Scientists in many countries show that ^{235}U is fissioned by slow neutrons, but ^{232}Th and ^{238}U only by fast neutrons, and that each fission produces two or three new neutrons while large amounts of energy are released. The possibility of producing nuclear weapons and building nuclear power stations is considered in several countries.
1942	**Fermi** and co-workers build the first nuclear reactor (critical on December 2).
1944	First gram amounts of a synthetic element (Pu) produced at Oak Ridge, USA. Kilogram quantities produced in Hanford, USA, in 1945.
1944	**McMillan** and **Veksler** discover the synchrotron principle which makes it possible to build accelerators for energies > 1000 MeV.
1940-1945	**Oppenheimer** and co-workers develop a device to produce fast uncontrolled chain reactions releasing very large amounts of energy. First test at Alamogordo, New Mexico, USA, on July 16, 1945 produces an energy corresponding to 20,000 tons of TNT; this is followed by the use of atomic bombs on Hiroshima (Aug. 6, 1945) and on Nagasaki (Aug. 9, 1945).
1944-1947	Photo-multiplier scintillation detectors are developed.
1946	**Libby** develops the ^{14}C-method for age determination.
1946	First Soviet nuclear reactor starts.
1949	Soviet tests a nuclear bomb.
1950	**Mayer, Haxel, Jensen** and **Suess** suggest a nuclear shell model.
1951	The first breeder reactor, which also produces the first electric power, is developed by Argonne National Laboratory, USA, and built in Idaho.
1952	The United States test the first device for uncontrolled large scale fusion power (the hydrogen bomb).
1953-1955	**A. Bohr, Mottelson,** and **Nilsson** develop the unified model of the nucleus (single particle effects on collective motions).
1955	**Chamberlain, Segré, Wiegand,** and **Ypsilantis** produce antiprotons.
1955	First nuclear powered ship (submarine *Nautilus*).
1954-1956	A 5 MW_e nuclear power station starts at Obninsk, USSR, in 1954. First civilian nuclear power station (45 MW_e) starts at Calder Hall, England, in 1956.
1956	**Reines** and **Cowan** prove the existence of neutrinos.
1957	Fire in carbon dioxide cooled graphite reactor at Windscale, U.K.
1957	Explosion in nuclear waste storage facility at Kyshtym (Chelyabinsk), USSR, with contamination of large areas.

(*Continued*)

Table 1.1 Historical survey of nuclear science—cont'd

1959	First civilian ship reactor used in the ice-breaker *Lenin*, launched in the USSR.
~1960	**Hofstadter** et al.; protons and neutrons contain unevenly distributed internal charge.
~1960	**Lederman, Schwarz** and **Steinberger** discover the muon neutrino.
1961	A radionuclide (^{238}Pu) is used as power source in a satellite (Transit-4 A).
1961	Semiconductor detectors are developed.
1963	End of atmospheric testing of nuclear weapons (see below).
1965	**A. Penzias** and **R. W. Wilson** discover the 3 K cosmic microwave radiation background.
~1970	Theory of quarks developed (**Gell-Mann**); quarks proven in nuclear scattering experiments (**Friedman, Kendall and Taylor**).
1972	French scientists discover ancient natural nuclear reactor in Oklo, Gabon.
1979	Core melt-down in PWR reactor at the Three Mile Island plant near Harrisburg, USA; no environmental contamination.
1983	**Rubbia, van der Meer** & co-workers at CERN discover the W and Z weak interaction communicators.
1986	Explosion and fire in the Chernobyl-4 reactor unit at Pripyat, Ukraine, USSR, with contamination of large areas.
2011	Fukushima Daichi nuclear power station in Japan is hit, and badly damaged, by a very large tsumani.
2013	Higgs boson discovered at CERN
International events concerning radiation and nuclear energy	
1955	Formation of United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR).
1957	Formation of the International Atomic Energy Agency (IAEA), with headquarters in Vienna.
1963	Partial Test Ban Treaty bans nuclear tests in the atmosphere, in outer space, and under water.
1968	Treaty on the Non-Proliferation of Nuclear Weapons (NPT) is signed by the "three depository governments" (USSR, UK, and USA), all nuclear weapons countries (NWC), and 40 other signatory, non-nuclear weapons countries (NNWC).
1971	The IAEA takes the responsibility for a safeguards system for control of fissile material in non-nuclear weapons countries.
1991	140 states have signed the NPT agreement.
2011	Germany decides to abandon nuclear energy production within a few years.

1.6. LITERATURE

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